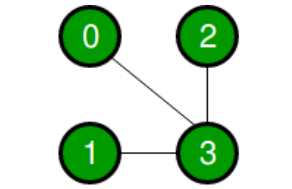
2-approximation algorithm for Vertex Cover problem

- A vertex cover of an undirected graph is a subset of its vertices such that for every edge (u, v) of the graph, either 'u' or 'v' is in the vertex cover.
- Although the name is Vertex Cover, the set covers all edges of the given graph.

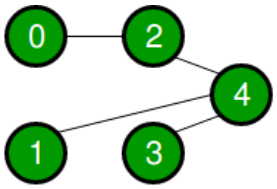
Given an undirected graph, the vertex cover problem is to find minimum size vertex cover.



Minimum vertex cover is empty{}



Minimum vertex cover is {3}



Minimum vertex cover is {4, 2} or {4, 0}

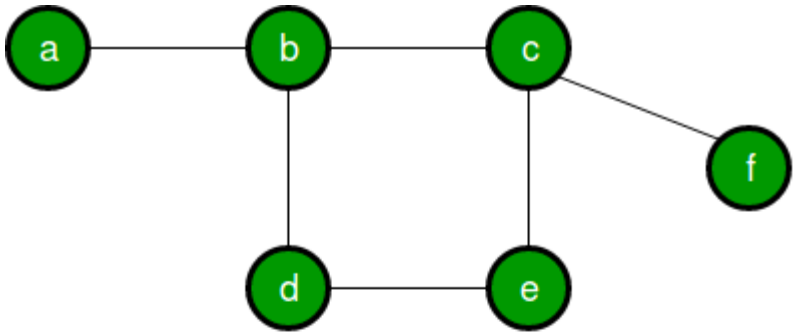
Approx-Vertex-Cover

- Vertex Cover can be approximated by the following simple algorithm, which iterately chooses an edge that is not covered yet and covers it:

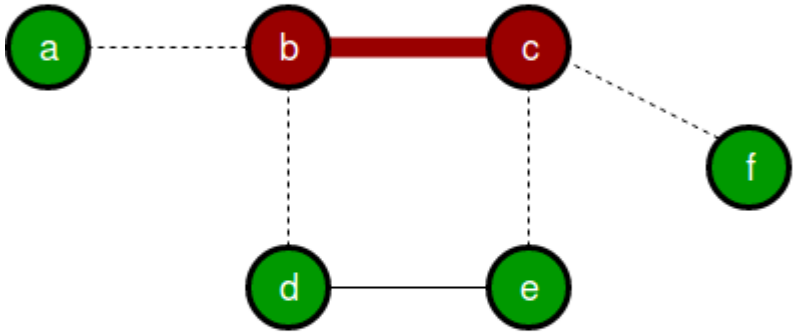
1) **Initialize the result as {}**
2) Consider a set of all edges in given graph. Let the set be E.
3) Do following while E is not empty
...a) Pick an arbitrary edge (u, v) from set E and add 'u' and 'v' to result
...b) Remove all edges from E which are either incident on u or v.
4) Return result

```
APPROX-VERTEX-COVER(G)
1  C = ∅
2  E' = G.E
3  while E' ≠ ∅
4      let (u, v) be an arbitrary edge of E'
5      C = C ∪ {u, v}
6      remove from E' every edge incident on either u or v
7  return C
```

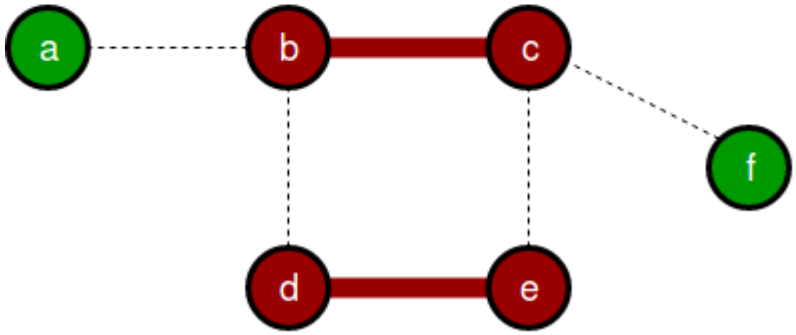
- Below diagram to show the execution of the above approximate algorithm:



(a)



(b)



(c)

Minimum Vertex Cover is {b, c, d} or {b, c, e}

How well the above algorithm perform?

- It can be proved that the above approximate algorithm never finds a vertex cover whose size is more than twice the size of the minimum possible vertex cover.

- **Theorem:** *APPROX-VERTEX-COVER is a polynomial-time 2-approximate algorithm i.e., the algorithm has a ration bound of 2.*
- **Goal:** Since this a minimization problem, we are interested in smallest possible c/c^*. Specifically we want to show $c/c^* \leq 2 = p(n)$. In other words, we want to show that APPROX-VERTEX-COVER algorithm returns a vertex-cover that is atmost twice the size of an optimal cover.
- ***Proof:*** Let the set c and c^* be the sets output by APPROX-VERTEX -COVER and OPTIMAL-VERTEX-COVER respectively. Also, let A be the set of edges selected by line 4.
- Because, we have added both vertices, we get $c = 2|A|$ but OPTIMAL-VERTEX-COVER would have add one of two.
- => $c/c^* \leq p(n) = 2$.
- Formally, since no two edge in A are covered by the same vertex from c^* (since, once an edge is picked in line 4, all other edges that are incident on its endpoints are deleted from E in line 6) and we the lower bound:
- $|c^*| \geq A$ ---------- 1
- On the size of an OPTIMAL-VERTEX-COVER.
- In line 4, we picked both en points yielding an upper bound on the size of Vertex-Cover.
- $|c| \leq 2|A|$
- Since, upper bound is an exact in this case, we have
- $|c| = 2|A|$ ---------- 2

  Take $|c|/2 = |A|$ and put it in equation 1

  $|c^*| \geq |c|/2$

  $|c^*|/|c| \geq 1/2$

  $|c^*|/|c| \leq 2 = p(n)$
- proving the theorem. □
-